An Agile View of Process

Key Concepts

agile manifesto
agile modeling
agile process models
agility
agility principles
ASD
Crystal
DSDM
Extreme Programming
FDD
pair programming
politics
refactoring
Scrum
team characteristics

In 2001, Kent Beck and 16 other noted software developers, writers, and consultants [BEC01a] (referred to as the "Agile Alliance") signed the "Manifesto for Agile Software Development." It stated:

> We are uncovering better ways of developing software by doing it and helping others do it. Through this work we have come to value:
>
> *Individuals and interactions* over processes and tools
>
> *Working software* over comprehensive documentation
>
> *Customer collaboration* over contract negotiation
>
> *Responding to change* over following a plan
>
> That is, while there is value in the items on the right, we value the items on the left more.

A manifesto is normally associated with an emerging political movement—one that attacks the old guard and suggests revolutionary change (hopefully for the better). In some ways, that's exactly what agile development is all about.

Although the underlying ideas that guide agile development have been with us for many years, it has only been during the past decade that these ideas have crystallized into a "movement." In essence, agile[1] methods were developed in an effort to overcome perceived and actual weaknesses in conventional software engineering. Agile development can provide important benefits, but it is not applicable to all projects, products, people, and situations. It is also *not*

Quick Look

What is it? Agile software engineering combines a philosophy and a set of development guidelines. The philosophy encourages customer satisfaction and early incremental delivery of software; small, highly motivated project teams; informal methods; minimal software engineering work products; and overall development simplicity. The development guidelines stress delivery over analysis and design (although these activities are not discouraged), and active and continuous communication between developers and customers.

Who does it? Software engineers and other project stakeholders (managers, customers, end-users) work together on an agile team—a team that is self-organizing and in control of its own destiny. An agile team fosters communication and collaboration among all who serve on it.

Why is it important? The modern business environment that spawns computer-based systems and software products is fast-paced and

1 Agile methods are sometimes referred to as *light* or *lean* methods.

ever-changing. Agile software engineering represents a reasonable alternative to conventional software engineering for certain classes of software and certain types of software projects. It has been demonstrated to deliver successful systems quickly.

What are the steps? Agile development might best be termed "software engineering lite." The basic framework activities—customer communication, planning, modeling, construction, delivery and evaluation—remain. But they morph into a minimal task set that pushes the project team toward construction and delivery (some would argue that this is done at the expense of problem analysis and solution design).

What is the work product? Customers and software engineers who have adopted the agile philosophy have the same view—the only really important work product is an operational "software increment" that is delivered to the customer on the appropriate commitment date.

How do I ensure that I've done it right? If the agile team agrees that the process works and the team produces deliverable software increments that satisfy the customer, you've done it right.

antithetical to solid software engineering practice and can be applied as an overriding philosophy for all software work.

In the modern economy, it is often difficult or impossible to predict how a computer-based system (e.g., a Web-based application) will evolve as time passes. Market conditions change rapidly, end-user needs evolve, and new competitive threats emerge without warning. In many situations, we no longer are able to define requirements fully before the project begins. Software engineers must be agile enough to respond to a fluid business environment.

Does this mean that a recognition of these modern realities causes us to discard valuable software engineering principles, concepts, methods, and tools? Absolutely not! Like all engineering disciplines, software engineering continues to evolve. It can be adapted easily to meet the challenges posed by a demand for agility.

> "Agility: 1, everything else: 0."
>
> Tom DeMarco

In a thought-provoking book on agile software development, Alistair Cockburn [COC02a] argues that the prescriptive process models introduced in Chapter 3 have a major failing: *they forget the frailties of the people who build computer software*. Software engineers are not robots. They exhibit great variation in working styles and significant differences in skill level, creativity, orderliness, consistency, and spontaneity. Some communicate well in written form, others do not. Cockburn argues that process models can "deal with people's common weaknesses with [either] discipline or tolerance" [COC02a] and that most prescriptive process models choose discipline. He states: "Because consistency in action is a human weakness, high discipline methodologies are fragile" [COC02a].

If process models are to work, they must provide a realistic mechanism for encouraging the discipline that is necessary, or they must be characterized in a manner that shows "tolerance" for the people who do software engineering work. Invariably, tolerant practices are easier for software people to adopt and sustain, but (as Cockburn admits) they may be less productive. Like most things in life, trade-offs must be considered.

4.1 What Is Agility?

Just what is agility in the context of software engineering work? Ivar Jacobson [JAC02] provides a useful discussion:

> *Agility* has become today's buzzword when describing a modern software process. Everyone is agile. An agile team is a nimble team able to appropriately respond to changes. Change is what software development is very much about. Changes in the software being built, changes to the team members, changes because of new technology, changes of all kinds that may have an impact on the product they build or the project that creates the product. Support for changes should be built-in everything we do in software, something we embrace because it is the heart and soul of software. An agile team recognizes that software is developed by individuals working in teams and that the skills of these people, their ability to collaborate is at the core for the success of the project.

In Jacobson's view, the pervasiveness of change is the primary driver for agility. Software engineers must be quick on their feet if they are to accommodate the rapid changes that Jacobson describes.

> Agility is dynamic, content specific, aggressively change embracing, and growth oriented.
>
> **Steven Goldman et al.**

ADVICE

Don't make the mistake of assuming that agility gives you license to hack out solutions. A process is required, and discipline is essential.

But agility is more than an effective response to change. It also encompasses the philosophy espoused in the manifesto noted at the beginning of this chapter. It encourages team structures and attitudes that make communication (among team members, between technologists and business people, between software engineers and their managers) more facile. It emphasizes rapid delivery of operational software and de-emphasizes the importance of intermediate work products (not always a good thing); it adopts the customer as a part of the development team and works to eliminate the "us and them" attitude that continues to pervade many software projects; it recognizes that planning in an uncertain world has its limits and that a project plan must be flexible.

The Agile Alliance [AGI03] defines 12 principles for those who want to achieve agility:

1. Our highest priority is to satisfy the customer through early and continuous delivery of valuable software.
2. Welcome changing requirements, even late in development. Agile processes harness change for the customer's competitive advantage.
3. Deliver working software frequently, from a couple of weeks to a couple of months, with a preference to the shorter timescale.

4. Business people and developers must work together daily throughout the project.

5. Build projects around motivated individuals. Give them the environment and support they need, and trust them to get the job done.

6. The most efficient and effective method of conveying information to and within a development team is face-to-face conversation.

7. Working software is the primary measure of progress.

8. Agile processes promote sustainable development. The sponsors, developers, and users should be able to maintain a constant pace indefinitely.

9. Continuous attention to technical excellence and good design enhances agility.

10. Simplicity—the art of maximizing the amount of work not done—is essential.

11. The best architectures, requirements, and designs emerge from self-organizing teams.

12. At regular intervals, the team reflects on how to become more effective, then tunes and adjusts its behavior accordingly.

Agility can be applied to any software process. However, to accomplish this, it is essential that the process be designed in a way that allows the project team to adapt tasks and to streamline them, conduct planning in a way that understands the fluidity of an agile development approach, eliminate all but the most essential work products and keep them lean, and emphasize an incremental delivery strategy that gets working software to the customer as rapidly as feasible for the product type and operational environment.

4.2 What Is an Agile Process?

Any *agile software process* is characterized in a manner that addresses three key assumptions [FOW02] about the majority of software projects:

1. It is difficult to predict in advance which software requirements will persist and which will change. It is equally difficult to predict how customer priorities will change as a project proceeds.

2. For many types of software, design and construction are interleaved. That is, both activities should be performed in tandem so that design models are proven as they are created. It is difficult to predict how much design is necessary before construction is used to prove the design.

3. Analysis, design, construction, and testing are not as predictable (from a planning point of view) as we might like.

WebRef
A comprehensive collection of articles on the agile process can be found at **www.aanpo.org/articles/index.**

Given these three assumptions, an important question arises: How do we create a process that can manage unpredictability? The answer, as we have already noted, lies in process adaptability (to rapidly changing project and technical conditions). An agile process, therefore, must be *adaptable.*

But continual adaptation without forward progress accomplishes little. Therefore, an agile software process must adapt *incrementally.* To accomplish incremental adaptation, an agile team requires customer feedback (so that the appropriate adaptations can be made). An effective catalyst for customer feedback is an operational prototype or a portion of an operational system. Hence, an *incremental development strategy* should be instituted. *Software increments* (executable prototypes or a portion of an operational system) must be delivered in short time periods so that adaptation keeps pace with change (unpredictability). This iterative approach enables the customer to evaluate the software increment regularly, provide necessary feedback to the software team, and influence the process adaptations that are made to accommodate the feedback.

> "There is no substitute for rapid feedback, both on the development process and on the product itself."
> Martin Fowler

4.2.1 The Politics of Agile Development

There is considerable debate (sometimes strident) about the benefits and applicability of agile software development as opposed to more conventional software engineering processes. Jim Highsmith [HIG02a] (facetiously) states the extremes when he characterizes the feeling of the pro-agility camp ("agilists"). "Traditional methodologists are a bunch of stick-in-the-muds who'd rather produce flawless documentation than a working system that meets business needs." As a counterpoint, he states (again, facetiously) the position of the traditional software engineering camp: "Lightweight, er, 'agile' methodologists are a bunch of glorified hackers who are going to be in for a heck of a surprise when they try to scale up their toys into enterprise-wide software."

Like all software technology arguments, this methodology debate risks degenerating into a religious war. If warfare breaks out, rational thought disappears and beliefs rather than facts guide decision-making.

No one is against agility. The real question is: What is the best way to achieve it? As important, how do we build software that meets customers' needs today and exhibits the quality characteristics that will enable it to be extended and scaled to meet customers' needs over the long term?

ADVICE
You don't have to choose between agility and software engineering. Instead, define a software engineering approach that is agile.

There are no absolute answers to either of these questions. Even within the agile school itself, there are many proposed process models (Section 4.3), each with a subtly different approach to the agility problem. Within each model there is a set of "ideas" (agilists are loath to call them "work tasks") that represent a significant departure from conventional software engineering. And yet, many agile concepts are simply adaptations of good software engineering concepts. Bottom line: there is much that can be gained by considering the best of both schools and virtually nothing to be gained by denigrating either approach.

The interested reader should see [HIG01], [HIG02a], and [DEM02] for an entertaining summary of the important technical and political issues.

4.2.2 Human Factors

Proponents of agile software development take great pains to emphasize the importance of "people factors" in successful agile development. As Cockburn and Highsmith [COC01] state, "Agile development focuses on the talents and skills of individuals, molding the process to specific people and teams." The key point in this statement is that the *process molds to the needs of the people and team,* not the other way around.[2]

> "What counts as barely sufficient for one team is either overly sufficient or insufficient for another."
>
> Alistair Cockburn

If members of the software team are to drive the characteristics of the process that is applied to build software, a number of key traits must exist among the people on an agile team and the team itself:

What key traits must exist among the people on an effective software team?

Competence. In an agile development (as well as conventional software engineering) context, "competence" encompasses innate talent, specific software related skills, and overall knowledge of the process that the team has chosen to apply. Skill and knowledge of process can and should be taught to all people who serve as agile team members.

Common focus. Although members of the agile team may perform different tasks and bring different skills to the project, all should be focused on one goal—to deliver a working software increment to the customer within the time promised. To achieve this goal, the team will also focus on continual adaptations (small and large) that will make the process fit the needs of the team.

Collaboration. Software engineering (regardless of process) is about assessing, analyzing, and using information that is communicated to the software team; creating information that will help the customer and others understand the work of the team; and building information (computer software and relevant databases) that provides business value for the customer. To accomplish these tasks, team members must collaborate—with one another, with the customer, and with business managers.

Decision-making ability. Any good software team (including agile teams) must be allowed the freedom to control its own destiny. This implies that the

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2 Most successful software engineering organizations recognize this reality regardless of the process model they choose.

team is given autonomy—decision-making authority for both technical and project issues.

Fuzzy problem-solving ability. Software managers should recognize that the agile team will continually have to deal with ambiguity and will continually be buffeted by change. In some cases, the team must accept the fact that the problem they are solving today may not be the problem that needs to be solved tomorrow. However, lessons learned from any problem solving activity (including those that solve the wrong problem) may be of benefit to the team later in the project.

Mutual trust and respect. The agile team must become what DeMarco and Lister [DEM98] call a "jelled" team (see Chapter 21). A jelled team exhibits the trust and respect that are necessary to make them "so strongly knit that the whole is greater than the sum of the parts" [DEM98].

POINT

A self-organizing team is in control of the work it performs. The team makes its own commitments and defines plans to achieve them.

Self-organization. In the context of agile development, *self-organization* implies three things: (1) the agile team organizes itself for the work to be done; (2) the team organizes the process to best accommodate its local environment; (3) the team organizes the work schedule to best achieve delivery of the software increment. Self-organization has a number of technical benefits, but more importantly it serves to improve collaboration and boost team morale. In essence, the team serves as its own management. Ken Schwaber [SCH02] addresses these issues when he writes: "The team selects how much work it believes it can perform within the iteration, and the team commits to the work. Nothing demotivates a team as much as someone else making commitments for it. Nothing motivates a team as much as accepting the responsibility for fulfilling commitments that it made itself."

4.3 Agile Process Models

The history of software engineering is littered with dozens of obsolete process descriptions and methodologies, modeling methods and notations, tools, and technology. Each flared in notoriety and was then eclipsed by something new and (purportedly) better. With the introduction of a wide array of agile process models—each contending for acceptance within the software development community—the agile movement is following the same historical path.[3]

> "Our profession goes through methodologies like a 14-year-old goes through clothing."
>
> Stephen Hawrysh and Jim Ruprecht

3 This is not a bad thing. Before one or more models or methods are accepted as a de facto standard, all must contend for the hearts and minds of software engineers. The "winners" evolve into best practice while the "losers" either disappear or merge with the winning models.

In the sections that follow, we present an overview of a number of different *agile process models.* There are many similarities (in philosophy and practice) among these approaches. Our intent will be to emphasize those characteristics of each method that make it unique. It is important to note that *all* agile models conform (to a greater or lesser degree) to the *Manifesto for Agile Software Development* and the principles noted in Section 4.1.

4.3.1 Extreme Programming (XP)

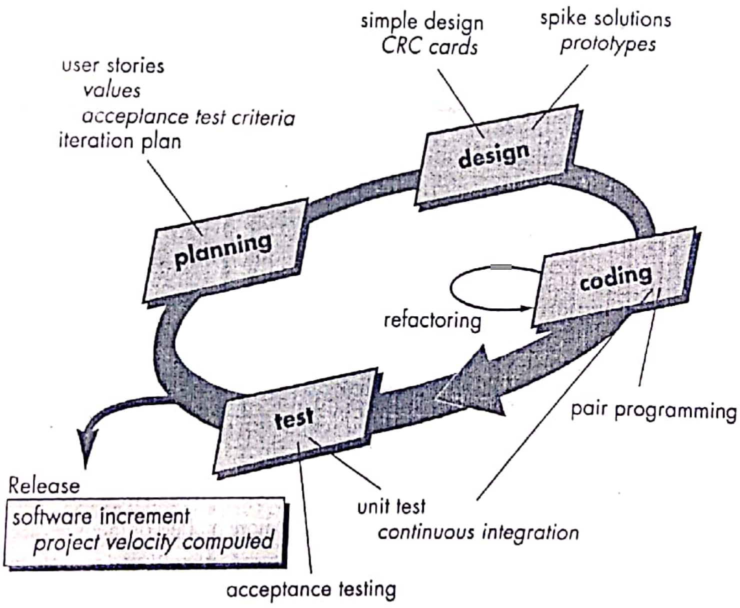
Although early work on the ideas and methods associated with *Extreme Programming (XP)* occurred during the late 1980s, the seminal work on the subject, written by Kent Beck [BEC99] was published in 1999. Subsequent books by Jeffries et al [JEF01] on the technical details of XP, and additional work by Beck and Fowler [BEC01b] on XP planning, flesh out the details of the method.

> **WebRef**
> An excellent overview of "rules" for XP can be found at **www.extremeprogramming.org/rules.html.**

XP uses an object-oriented approach (Part 2 of this book) as its preferred development paradigm. XP encompasses a set of rules and practices that occur within the context of four framework activities: planning, design, coding, and testing. Figure 4.1 illustrates the XP process and notes some of the key ideas and tasks that are associated with each framework activity. Key XP activities are summarized in the paragraphs that follow.

Planning. The planning activity begins with the creation of a set of *stories* (also called *user stories*) that describe required features and functionality for software to be built. Each story (similar to use-cases described in Chapters 7 and 8) is written by the customer and is placed on an index card. The customer assigns a *value* (i.e., a

FIGURE 4.1
The Extreme Programming process



Code Refactoring: is the process of restructuring existing computer code– changing the factoring– without changing its external behaviour. Refactoring is intended to improve nonfunctional attributes of the software

What is an XP "story"?

priority) to the story based on the overall business value of the feature or function.[4] Members of the XP team then assess each story and assign a *cost*—measured in development weeks—to it. If the story will require more than three development weeks, the customer is asked to split the story into smaller stories, and the assignment of value and cost occurs again. It is important to note that new stories can be written at any time.

Customers and the XP team work together to decide how to group stories into the next release (the next software increment) to be developed by the XP team. Once a basic *commitment* (agreement on stories to be included, delivery date, and other project matters) is made for a release, the XP team orders the stories that will be developed in one of three ways: (1) all stories will be implemented immediately (within a few weeks); (2) the stories with highest value will be moved up in the schedule and implemented first; or (3) the riskiest stories will be moved up in the schedule and implemented first.

WebRef
A worthwile XP "planning game" can be found at c2.com/cgi/wiki?planningGame.

After the first project release (also called a software increment) has been delivered, the XP team computes project velocity. Stated simply, *project velocity* is the number of customer stories implemented during the first release. Project velocity can then be used to (1) help estimate delivery dates and schedule for subsequent releases, and (2) determine whether an over-commitment has been made for all stories across the entire development project. If an over-commitment occurs, the content of releases is modified or end-delivery dates are changed.

As development work proceeds, the customer can add stories, change the value of an existing story, split stories, or eliminate them. The XP team then reconsiders all remaining releases and modifies its plans accordingly.

> "Extreme Programming is a discipline of software development based on values of simplicity, communication, feedback, and courage."
>
> **Ron Jeffries**

Design. XP design rigorously follows the KIS (keep it simple) principle. A simple design is always preferred over a more complex representation. In addition, the design provides implementation guidance for a story as it is written—nothing less, nothing more. The design of extra functionality (because the developer assumes it will be required later) is discouraged.[5]

XP encourages the use of CRC cards (Chapter 8) as an effective mechanism for thinking about the software in an object-oriented context. CRC (class-responsibility collaborator) cards identify and organize the object-oriented classes[6] that are relevant to the current software increment. The XP team conducts the design exercise using a

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4 The value of a story may also depend on the presence of another story.

5 These design guidelines should be followed in every software engineering method, although there are times when sophisticated design notation and terminology may get in the way of simplicity.

6 Object-oriented classes are discussed in detail in Chapter 8 and throughout Part 2 of this book.

process similar to the one described in Chapter 8 (Section 8.7.4). The CRC cards are the only design work product produced as part of the XP process.

If a difficult design problem is encountered as part of the design of a story, XP recommends the immediate creation of an operational prototype of that portion of the design. Called a *spike solution,* the design prototype is implemented and evaluated. The intent is to lower risk when true implementation starts and to validate the original estimates for the story containing the design problem.

WebRef Refactoring techniques and tools can be found at **www.refactoring.com.**

XP encourages *refactoring*—a construction technique that is also a design technique. Fowler [FOW00] describes refactoring in the following manner:

> Refactoring is the process of changing a software system in such a way that it does not alter the external behavior of the code yet improves the internal structure. It is a disciplined way to clean up code [and modify/simplify the internal design] that minimizes the chances of introducing bugs. In essence, when you refactor you are improving the design of the code after it has been written.

Because XP design uses virtually no notation and produces few, if any work products other than CRC cards and spike solutions, design is viewed as a transient artifact that can and should be continually modified as construction proceeds. The intent of refactoring is to control these modifications by suggesting small design changes that "can radically improve the design" [FOW00]. It should be noted, however, that effort required for refactoring can grow dramatically as the size of an application grows.

A central notion in XP is that design occurs both before *and after* coding commences. Refactoring means that design occurs continuously as the system is constructed. In fact, the construction activity itself will provide the XP team with guidance on how to improve the design.

WebRef Useful information on XP can be obtained at **www.xprogramming.com.**

Coding. XP recommends that after stories are developed and preliminary design work is done, the team should not move to code, but rather develop a series of unit tests that will exercise each of the stories that is to be included in the current release (software increment).[7] Once the unit test has been created, the developer is better able to focus on what must be implemented to pass the unit test. Nothing extraneous is added (KIS). Once the code is complete, it can be unit tested immediately, thereby providing instantaneous feedback to the developers.

What is pair programming?

A key concept during the coding activity (and one of the most talked about aspects of XP) is *pair programming.* XP recommends that two people work together at one computer workstation to create code for a story. This provides a mechanism for real-time problem solving (two heads are often better than one) and real-time quality assurance. It also keeps the developers focused on the problem at hand. In practice, each person takes on a slightly different role. For example, one person might think about the coding details of a particular portion of the design while the other ensures

7 This approach is analogous to knowing the exam questions before you begin to study. It makes studying much easier by focusing attention only on the questions that will be asked.

that coding standards (a required part of XP) are being followed and the code that is generated will "fit" into the broader design for the story.

As pair programmers complete their work, the code they develop is integrated with the work of others. In some cases this is performed on a daily basis by an integration team. In other cases, the pair programmers have integration responsibility. This "continuous integration" strategy helps to avoid compatibility and interfacing problems and provides a "smoke testing" environment (Chapter 13) that helps to uncover errors early.

Testing. We have already noted that the creation of a unit test[8] before coding commences is a key element of the XP approach. The unit tests that are created should be implemented using a framework that enables them to be automated (hence, they can be executed easily and repeatedly). This encourages a regression testing strategy (Chapter 13) whenever code is modified (which is often, given the XP refactoring philosophy).

As the individual unit tests are organized into a "universal testing suite" [WEL99], integration and validation testing of the system can occur on a daily basis. This provides the XP team with a continual indication of progress and also can raise warning flags early if things are going awry. Wells [WEL99] states: "Fixing small problems every few hours takes less time than fixing huge problems just before the deadline."

KEY POINT
XP acceptance tests are derived from user stories.

XP *acceptance tests*, also called *customer tests*, are specified by the customer and focus on overall system features and functionality that are visible and reviewable by the customer. Acceptance tests are derived from user stories that have been implemented as part of a software release.

SAFEHOME

Considering Agile Software Development

The scene: Doug Miller's office.

The players: Doug Miller, software engineering manager; Jamie Lazar, software team member; Vinod Raman, software team member.

The conversation:
(A knock on the door)

Jamie: Doug, you got a minute?

Doug: Sure Jamie, what's up?

Jamie: We've been thinking about our process discussion yesterday . . . you know, what process we're going to choose for this new *SafeHome* project.

Doug: And?

Vinod: I was talking to a friend at another company, and he was telling me about Extreme Programming. It's an agile process model, heard of it?

Doug: Yeah, some good, some bad.

Jamie: Well, it sounds pretty good to us. Lets you develop software really fast, uses something called pair programming to do real-time quality checks . . . it's pretty cool, I think.

Doug: It does have a lot of really good ideas. I like the pair programming concept, for instance, and the idea that stakeholders should be part of the team.

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8 Unit testing, discussed in detail in Chapter 13, focuses on an individual software component, exercising the component's interface, data structures, and functionality in an effort to uncover errors that are local to the component.

Jamie: Huh? You mean that marketing will work on the project team with us?

Doug (nodding): They're a stakeholder, aren't they?

Jamie: Jeez . . . they'll be requesting changes every five minutes.

Vinod: Not necessarily. My friend said that there are ways to "embrace" changes during an XP project.

Doug: So you guys think we should use XP?

Jamie: It's definitely worth considering.

Doug I agree. And even if we choose an incremental model as our approach, there's no reason why we can't incorporate much of what XP has to offer.

Vinod: Doug, before you said "some good, some bad." What was the "bad"?

Doug: The thing I don't like is the way XP downplays analysis and design . . . sort of says that writing code is where the action is.

(The team members look at one another and smile.)

Doug: So you agree with the XP approach?

Jamie (speaking for both): Writing code is what we do, Boss!

Doug (laughing): True, but I'd like to see you spend a little less time coding and then re-coding and a little more time analyzing what has to be done and designing a solution that works.

Vinod: Maybe we can have it both ways, agility with a little discipline.

Doug: I think we can, Vinod. In fact, I'm sure of it.

4.3.2 Adaptive Software Development (ASD)

Adaptive Software Development (ASD) has been proposed by Jim Highsmith [HIG00] as a technique for building complex software and systems. The philosophical underpinnings of ASD focus on human collaboration and team self-organization. Highsmith [HIG98] discusses this when he writes:

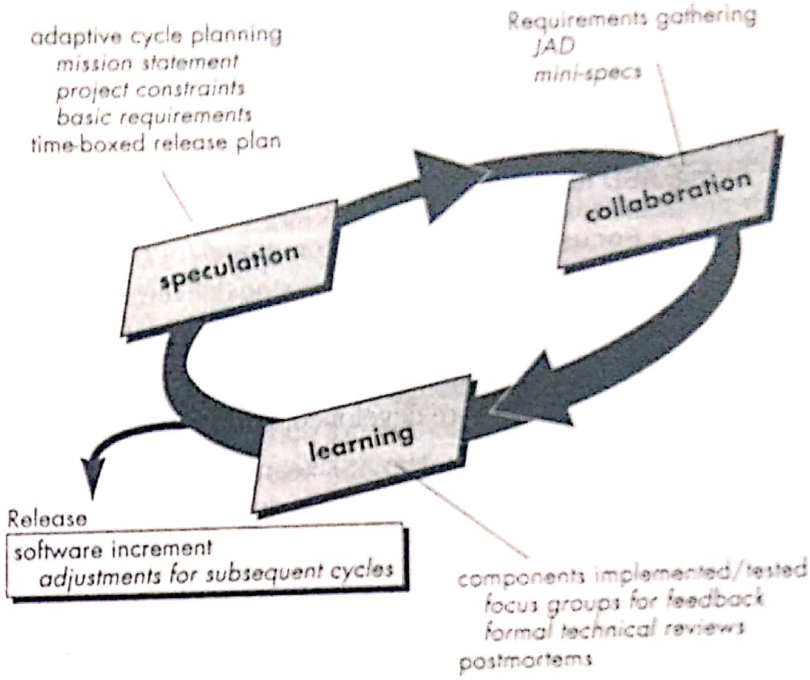
> Self-organization is a property of complex adaptive systems similar to a collective "aha," that moment of creative energy when the solution to some nagging problem emerges. Self-organization arises when individual, independent agents (cells in a body, species in an ecosystem, developers in a feature team) cooperate [collaborate] to create emergent outcomes. An emergent outcome is a property beyond the capability of any individual agent. For example, individual neurons in the brain do not possess consciousness, but collectively the property of consciousness emerges. We tend to view this phenomena of collective emergence as accidental, or at least unruly and undependable. The study of self-organization is proving that view to be wrong.

WebRef
Useful resources for ASD can be found at **www.adaptivesd.com.**

Highsmith argues that an agile, adaptive development approach based on collaboration is "as much a source of *order* in our complex interactions as discipline and engineering." He defines an ASD "life cycle" (Figure 4.2) that incorporates three phases: speculation, collaboration, and learning.

Speculation. During *speculation,* the project is initiated and *adaptive cycle planning* is conducted. Adaptive cycle planning uses project initiation information—the customer's mission statement, project constraints (e.g., delivery dates or user descrip-

FIGURE 4.2
Adaptive software development



What are the characteristics of ASD adaptive cycles?

ADVICE

Effective collaboration with your customer will only occur if you jettison any "us and them" attitudes.

tions), and basic requirements—to define the set of release cycles (software increments) that will be required for the project.[9]

Collaboration. Motivated people work together in a way that multiplies their talent and creative output beyond their absolute numbers. This collaborative approach is a recurring theme in all agile methods. But collaboration is not easy. It is not simply communication, although communication is a part of it. It is not only a matter of teamwork, although a "jelled" team (Chapter 21) is essential for real collaboration to occur. It is not a rejection of individualism, because individual creativity plays an important role in collaborative thinking. It is, above all, a matter of trust. People working together must trust one another to (1) criticize without animosity; (2) assist without resentment; (3) work as hard or harder as they do; (4) have the skill set to contribute to the work at hand; and (5) communicate problems or concerns in a way that leads to effective action.

> **"I like to listen. I have learned a great deal from listening carefully. Most people never listen."**
> **Ernest Hemingway**

9 Note that the adaptive cycle plan can and probably will be adapted to changing project and business conditions.

Learning. As members of an ASD team begin to develop the components that are part of an adaptive cycle, the emphasis is on learning as much as it is on progress toward a completed cycle. In fact, Highsmith [HIG00] argues that software developers often overestimate their own understanding (of the technology, the process, and the project) and that learning will help them to improve their level of real understanding. ASD teams learn in three ways:

1. **Focus groups.** The customer and/or end-users provide feedback on software increments that are being delivered. This provides a direct indication of whether or not the product is satisfying business needs.
2. **Formal technical reviews.** ASD team members review the software components that are developed, improving quality and learning as they proceed.
3. **Postmortems.** The ASD team becomes introspective, addressing its own performance and process (with the intent of learning and then improving its approach).

It is important to note that the ASD philosophy has merit regardless of the process model that is used. ASD's overall emphasis on the dynamics of self-organizing teams, interpersonal collaboration, and individual and team learning yield software project teams that have a much higher likelihood of success.

4.3.4 Scrum

Scrum (the name derived from an activity[10] that occurs during a rugby match) is an agile process model that was developed by Jeff Sutherland and his team in the early 1990s. In recent years, further development of the Scrum methods has been performed by Schwaber and Beedle [SCH01]. Scrum principles [ADM96] are consistent with the agile manifesto:

- Small working teams are organized to "maximize communication, minimize overhead, and maximize sharing of tacit, informal knowledge."
- The process must be adaptable to both technical and business changes "to ensure the best possible product is produced."
- The process yields frequent software increments "that can be inspected, adjusted, tested, documented, and built on."
- Development work and the people who perform it are partitioned "into clean, low coupling partitions, or packets."
- Constant testing and documentation is performed as the product is built.

10 A group of players forms around the ball and the teammates work together (sometimes violently!) to move the ball downfield.

- The Scrum process provides the "ability to declare a product 'done' whenever required (because the competition just shipped, because the company needs the cash, because the user/customer needs the functions, because that was when it was promised . . ." [ADM96].

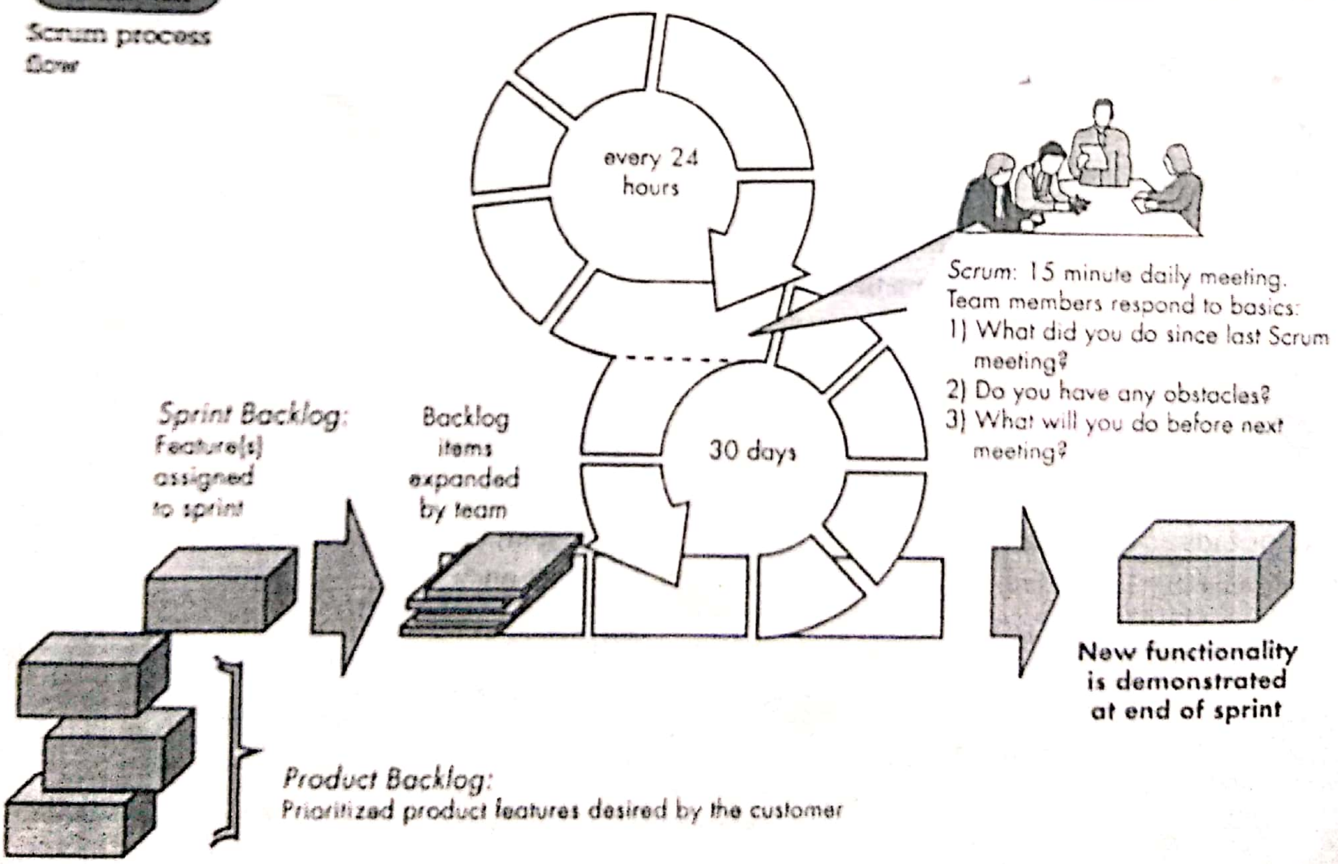
WebRef
Useful Scrum information and resources can be found at www.controlchaos.com.

Scrum principles are used to guide development activities within a process that incorporates the following framework activities: requirements, analysis, design, evolution, and delivery. Within each framework activity, work tasks occur within a process pattern (discussed in the following paragraph) called a *sprint*. The work conducted within a sprint (the number of sprints required for each framework activity will vary depending on product complexity and size) is adapted to the problem at hand and is defined and often modified in real-time by the Scrum team. The overall flow of the Scrum process is illustrated in Figure 4.3.

> "Scrum allows us to build softer software."
>
> **Mike Beetle et al.**

Scrum emphasizes the use of a set of "software process patterns" [NOY02] that have proven effective for projects with tight timelines, changing requirements, and business criticality. Each of these process patterns defines a set of development activities.

FIGURE 4.3
Scrum process flow



POINT

Scrum incorporates a set of process patterns that emphasize project priorities, compartmentalized work units, communication, and frequent customer feedback.

Backlog—a prioritized list of project requirements or features that provide business value for the customer. Items can be added to the backlog at any time (this is how changes are introduced). The product manager assesses the backlog and updates priorities as required.

Sprints—consist of work units that are required to achieve a requirement defined in the backlog that must be fit into a predefined time-box (typically 30 days). During the sprint, the backlog items that the sprint work units address are frozen (i.e., changes are not introduced during the sprint). Hence, the sprint allows team members to work in a short-term, but stable environment.

Scrum meetings—are short (typically 15 minutes) meetings held daily by the Scrum team. Three key questions are asked and answered by all team members [NOY02]:

- What did you do since the last team meeting?
- What obstacles are you encountering?
- What do you plan to accomplish by the next team meeting?

A team leader, called a "Scrum master," leads the meeting and assesses the responses from each person. The Scrum meeting helps the team to uncover potential problems as early as possible. Also, these daily meetings lead to "knowledge socialization" [BEE99] and thereby promote a self-organizing team structure.

Demos—deliver the software increment to the customer so that functionality that has been implemented can be demonstrated and evaluated by the customer. It is important to note that the demo may not contain all planned functionality, but rather those functions that can be delivered within the time-box that was established.

Beedle and his colleagues [BEE99] present a comprehensive discussion of these patterns in which they state: "SCRUM assumes up-front the existence of chaos. . . ." The Scrum process patterns enable a software development team to work successfully in a world where the elimination of uncertainty is impossible.